

Contact

ravis.du.mca10@gmail.com

www.linkedin.com/in/ravi-shankar-k-45a77224 (LinkedIn)

Top Skills

LangGraph

LangSmith

React Native

Languages

English (Professional Working)

Hindi (Native or Bilingual)

Certifications

Building Your first RAG System using LlamaIndex

Machine Learning with Python: Foundations

Quantum Explorer 2023: Advanced Understanding Quantum Computing
Quantum Computing Fundamentals

Honors-Awards

UGC NET

Cheerboard Nomination for July 2014

Cheerboard Nomination for September 2023

Cheerboard Nomination for February 2023

Ravi Shankar Kushwha

GenAI | AI/ML | AWS | iOS | Quantum Computing Certified @IITM
West Delhi, Delhi, India

Summary

With 12+ years of experience in AI/ML, Generative AI, NLP, Computer Vision, and iOS development, I specialized in building intelligent, high-performance applications that integrate cutting-edge AI with cloud technologies, React, Python(FASTAPI) and Productionizing GenAI projects on AWS. My expertise spans machine learning, deep learning, predictive analytics, and full-cycle iOS app development, enabling businesses to leverage data-driven insights, automation, and intelligent user experiences. AI/ML & Generative AI ExpertiseExpertise in LLMs, ChatGPT APIs, LangChain, LlamaIndex, and Retrieval-Augmented Generation (RAG) for building conversational AI and intelligent automation solutions.Strong foundation in transformers, attention mechanisms, and finetuning LLMs to develop domain-specific AI models.Experience in vector databases, semantic search, and scalable AI deployment for enhanced information retrieval and personalization. iOS Development & AI IntegrationLead the development of high-performance iOS applications, integrating OpenTok (WebRTC), Core ML, Vision, and on-device AI models to enhance real-time decision-making.Skilled in architecting scalable mobile applications with Generative AI-powered assistants, recommendation engines, and real-time interaction features.Experience in developing AI-driven SDKs, deploying cloud-based AI services, and optimizing AI inference on mobile devices. Technology & Skills AI/ML & Generative AI: ChatGPT APIs, LangChain, LlamaIndex, Hugging Face, Vector Databases, RAG, Fine-tuning LLMs NLP & Computer Vision: RNN, CNN, LSTM, GRU, OpenCV, spaCy, NLTK, Gensim, Transformers, Attention Mechanisms Deep Learning & Model Deployment: TensorFlow, Keras iOS Development: Swift, SwiftUI, Combine, Objective-C, Core ML, OpenTok (WebRTC), SQLite, Core Data Architectures & Agile: MVC, MVVM, CI/CD (Jenkins), Agile (Scrum) Certifications & Passion Quantum Computing Certified (IITM) | Passionate about merging Generative

AI, mobile development, and cloud technologies to build next-gen AI applications.

Experience

Nagarro

13 years 5 months

Staff Engineer

October 2021 - Present (4 years 8 months)

Gurugram, Haryana, India

* Sentiment Analysis of Movie Reviews:

Built a sentiment analysis model using the IMDB dataset to classify reviews as positive or negative.

Utilized word embeddings (Word2Vec, GloVe) and deep learning (TensorFlow/Keras) to improve model accuracy.

Technologies: Gensim, NumPy, pandas, Matplotlib, sklearn, TensorFlow, Keras

* Smart TV Gesture Recognition:

Developed a gesture recognition system to control smart TVs via hand gestures, using deep learning models and a webcam.

Recognized five gestures and mapped them to TV commands using TensorFlow and Keras.

Technologies: TensorFlow, Keras, OpenCV, NumPy, pandas, Matplotlib, glob

* Melanoma Detection:

Created a CNN-based model to detect melanoma from images, aiding dermatologists in early-stage cancer detection.

Implemented techniques for image augmentation and classification using TensorFlow/Keras.

Technologies: TensorFlow, Keras, PIL, Augmentor, Matplotlib, pandas, NumPy

* Bike Sharing Case Study:

Analyzed bike demand factors using data from BoomBikes to improve revenue post-pandemic.

Performed statistical analysis to identify key drivers of bike demand and optimize business strategy.

Technologies: sklearn, Statsmodels, NumPy, pandas, Matplotlib, Seaborn

* Lending Club Case Study:

Identified risky loan applicants in the Lending Club dataset to reduce credit loss.

Analyzed data to uncover key factors driving loan default, contributing to better risk assessment strategies.

Technologies: NumPy, pandas, Matplotlib, Seaborn

* Working as a team lead on multiple iOS project at same time.

* Suggested new improvements and ideas for the current iOS apps.

* Estimation of new features of current iOS apps and new iOS apps.

* Developed SDK to use in multiple iOS apps.

Associate Lead (Technology)

April 2018 - October 2021 (3 years 7 months)

Gurgaon, India

* Working as team lead to successfully deliver iOS app as per client requirement.

* Working on video calling iOS app using OpenTok(WebRTC) .

* Successfully implemented finite state machine in video calling iOS app.

* Working experience in agile (Scrum) methodology environment.

Senior Software Engineer

April 2016 - March 2018 (2 years)

Gurgaon, India

*Working as a member of COE Mobility Team.

*Design Project architecture and Research on new topic and create POC as per client requirement.

*Prepare Estimation sheet for new project (WBS - work break down structure)

*Prepare Business solution document and Code review.

Software Engineer

July 2013 - March 2016 (2 years 9 months)

Gurgaon

* Responsible for iOS Mobile App and Framework development.

* Responsible for Image enhancement (Binarization) to extract all information from Image using OpenCV . These information are present in textual format on Image.

* Responsible for OCR engine(Tesseract) integration in iOS App and enhancement of result and speedup of OCRing using Tesseract.

- * Responsible for CLucene framework integration in iOS App to learn(Create Indexes) documents and get appropriate document from created Indexes of documents.
- * Worked to Implement Live Camera. Live Camera detect largest quadrilateral on Camera and crop that largest quadrilateral using iOS API and OpenCV API.
- * Worked on Image Processing using two libraries: OpenCV and Image Magick.

Software Engineer - Trainee
January 2013 - June 2013 (6 months)

Delhi University
Student
July 2010 - June 2013 (3 years)
MCA

Education

Indian Institute of Technology, Madras
Introduction to Quantum Computing and Qiskit, Quantum Computing · (July 2023 - May 2024)

International Institute of Information Technology Bangalore
Executive Programme in Machine Learning and Artificial Intelligence, Machine Learning and Artificial Intelligence · (May 2024 - June 2025)

Delhi University
Master of Computer Applications (MCA), Computer Science · (2010 - 2013)

Integral University, Lucknow, Uttar Pradesh
BCA, Computer Science · (2005 - 2008)